LEARNER GUIDE

Finding a side — when the unknown is in the denominator

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Last lesson you found a side by writing the ratio, substituting, rearranging and evaluating. This lesson is the same four lines — and still no rule to memorise — with one new thing to watch. Sometimes, when you substitute, the unknown x lands in the DENOMINATOR of the ratio (the bottom of the fraction). That happens whenever you are finding the hypotenuse, or the adjacent from the opposite. When x is in the denominator, the algebra takes an extra step: multiply both sides by x, then divide by the ratio, so the answer is a DIVISION — x = known ÷ ratio — not a multiplication. The whole point is that you do not learn a second rule: you just look at where x is in your equation and let the algebra tell you what to do. The question you are exploring is: what changes when the unknown is in the denominator?

What you are learning to notice

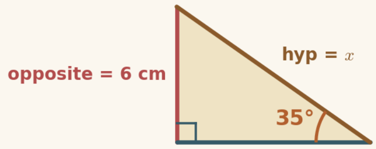
- That the unknown lands in the denominator whenever you are finding the hypotenuse (with sin or cos), or the adjacent from the opposite (with tan).
- That when x is in the denominator, you multiply both sides by x, then divide by the ratio: x = known ÷ ratio.
- That the position of x in the equation — numerator or denominator — tells you whether to multiply or divide.
- That there is no new rule: it is the same four lines, read from the equation you wrote.

Moving through it, one step at a time

In Numerator or denominator?, choose the ratio and reveal the working one line at a time — watch the two-step rearrangement (× by x, then ÷ by the ratio) unfold. In Side Finder (rearrange), do it yourself: choose the ratio, then type the rearrangement that gets x on its own (here you will divide), then evaluate on the calculator to 1 decimal place. In Multiply or divide?, look at where x sits in each equation and pick x = known × ratio or x = known ÷ ratio. In Reach the unmeasurable (the long way), find a sloping length you cannot reach — a ladder, a wire, a ramp. Keep your calculator in degrees (the ■ Calculator button is bottom-right), and write out every line.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

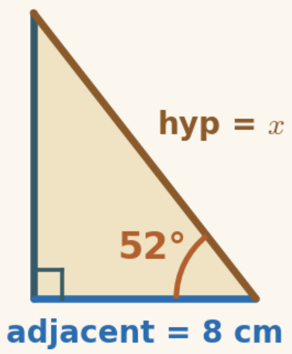


Example 1 — the unknown in the denominator (sine)

Here the opposite side is known (6 cm) and the hypotenuse is the unknown. When you write the ratio, x ends up in the denominator — so you divide.

1. write the ratio — $\sin 35° = \dfrac{\text{opposite}}{\text{hypotenuse}}$
2. put in the numbers — $\sin 35° = \dfrac{6}{x}$
3. rearrange (x is in the denominator, so divide) — $x = 6 \div \sin 35°$
4. work it out — $x = 6 \div 0.574$

x = 10.46 cm



Example 2 — the unknown in the denominator (cosine)

This time the adjacent is known (8 cm) and the hypotenuse is the unknown. Again x is in the denominator, so you divide.

1. write the ratio — $\cos 52° = \dfrac{\text{adjacent}}{\text{hypotenuse}}$
2. put in the numbers — $\cos 52° = \dfrac{8}{x}$
3. rearrange (x is in the denominator, so divide) — $x = 8 \div \cos 52°$
4. work it out — $x = 8 \div 0.616$

x = 12.99 cm

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (finding a side when the unknown is in the numerator (Lesson 5), the three ratios and SOH · CAH · TOA (Lesson 4), and rearranging an equation where the unknown is in the denominator (multiply both sides by x, then divide)), open a short recap and return whenever you are ready.

- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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